

AI-Powered Insurance Claims Assistant

Overview:

Insurance companies process large volumes of claims involving auto, home, renters, property, and business insurance. Claims teams need to review historical claims, identify similar cases, assess claim characteristics, detect potentially fraudulent patterns, prioritize claims, and understand factors affecting claim outcomes and resolution time. Manually reviewing large numbers of historical claims can be time-consuming and inconsistent.

Example Professional Questions:

- Has a similar claim been processed previously, and what was its outcome?
- What claim characteristics are associated with higher claim amounts or longer resolution times?
- Does a new claim show characteristics similar to previously flagged fraudulent claims?

Instead of manually searching through historical claim records, build an AI-powered assistant that can retrieve similar claims, analyze claim characteristics, identify suspicious patterns, and support claims-handling decisions.

Build a semantic retrieval and AI-assisted claims analysis system that can help insurance professionals investigate, assess, and prioritize claims.

Requirement 1 – Basic

- Build a Basic RAG system for historical claim similarity search
- Implement hybrid search using keyword and semantic retrieval
- Build a claim classification agent
- Implement metadata filtering based on claim type, state, claim status, claim amount, and policyholder claim history
- Generate basic claim assessment and handling recommendations
- Implement input validation guardrails
- Implement claim prioritization based on claim characteristics and risk indicators
- Expose the core functionality through API endpoints.

Requirement 2 – Advanced

- Implement DeepEval for evaluating claim-analysis and recommendation quality
- Build claim amount and resolution-time prediction models
- Implement LLM-as-Judge for validating claim analysis and recommendations
- Build a multi-agent insurance claims workflow for investigation, risk assessment, and recommendation
- Implement fraudulent-claim risk prediction

- Perform explainable analysis of claim characteristics and their impact on claim outcomes
- Implement a feedback loop for continuous improvement
- Build a simple front-end interface.

Deliverables:

1. **Architecture Diagram**
2. **Design**
3. **Full Executable Code (Microservice)**
4. **Panel Presentation (10 minutes)**

Dataset Name:

Free Synthetic Insurance Claims 100M

Primary Link:

[Free Synthetic Insurance Claims 100M – Hugging Face](#)

Alternative Link:

[Insurance Claims Fraud Data – Kaggle](#)

Format:

Parquet

Key Fields:

claim_id, policy_id, claim_type, state, policyholder_tenure_years, previous_claims_count, incident_date, claim_filed_date, claim_amount, deductible, claim_status, days_to_resolution, is_fraud_flagged_ground_truth

Reason:

The dataset contains **100 million synthetic property and casualty insurance claims** covering auto, home, renters, property, and business insurance. It includes claim characteristics, policyholder history, claim amounts, claim status, resolution time, and a ground-truth fraud indicator. These fields directly support claim similarity retrieval, hybrid search, classification, metadata filtering, claim prioritization, claim amount and resolution-time prediction, fraud-risk prediction, explainability, and AI-assisted claims recommendations.